AnsyTruss-2D User Manual

MATLAB-based 2D Truss Analysis Tool

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1 Introduction

AnsyTruss-2D is a **MATLAB-based** graphical user interface (GUI) app developed for the static analysis of two-dimensional pin-jointed trusses. The app uses the direct stiffness method to verify and analyse truss models defined by the user. Through editable input tables, users can define the truss geometry, material and sectional properties, support conditions, and applied nodal loads. The app then calculates nodal displacements, member axial forces, and support reactions, and provides visual outputs showing both the verified geometry and the deformed truss shape.

2 Coordinate System and Sign Convention

AnsyTruss-2D uses a global Cartesian coordinate system. The positive X-direction is defined from left to right, and the positive Y-direction is defined from bottom to top. Accordingly, nodal coordinates, applied loads, displacements, and support reactions should be interpreted using this sign convention.

3 App Panels

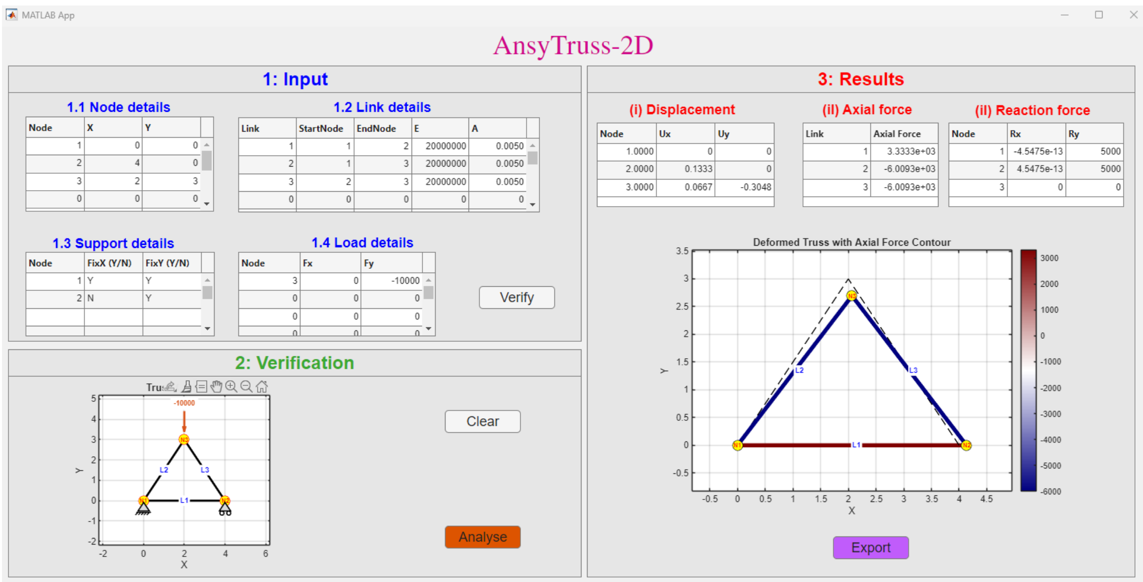

Figure 1: Overview of the AnsyTruss-2D graphical user interface showing the main app panels.

The GUI is organised into three main panels: the **Input Panel**, the **Output Panel**, and the **Visualisation/Verification Panel**. These panels are arranged to provide a clear workflow for defining the truss model, verifying the entered geometry and boundary conditions, and reviewing the analysis results.

3.1 Input Panel

The Input Panel is used to define the truss model. It allows the user to enter the required model data, including node coordinates, link connectivity, support conditions, and applied loads. The four main components are:

- Node Data
- Link Data
- Support Data
- Load Data

Users can type the required values directly into the tables. Blank rows may be left unused.

3.2 Verification Panel

The Verification Panel is used to check the truss model before analysis. After clicking the **Verify** button, the app plots the initial truss geometry and displays the entered model visually.

The verification plot displays:

- Node numbers
- Link numbers
- Pinned and roller support symbols
- Applied loads with appropriate directional arrows

This step helps confirm that the geometry, connectivity, support conditions, and applied loads have been entered correctly before running the analysis.

The **Clear** button is used to clear the entered input data from the input tables. This allows the user to reset the model definition and enter a new truss configuration without manually deleting each table entry.

3.3 Results Panel

The Results Panel displays the outputs of the truss analysis. After clicking the **Analyse** button, the app shows:

- Nodal displacement table
- Member axial force table

- Reaction force table
- Deformed truss shape overlaid on the initial undeformed geometry, with axial force contours displayed along the members

The **Export** button is provided to generate and save the analysis report in PDF format. The exported report includes the verified truss geometry, input data, deformed truss shape, displacement results, axial force results, and reactions.

4 Preloaded Example and Input Data Format

The preloaded example is a simple three-node triangular truss. This example is useful for quickly testing the **Verify**, **Analyse**, and **Export PDF Report** functions.

4.1 Node Data

The Node Data table defines the position of each node.

Node	X	Y
1	0.0	0.0
2	4.0	0.0
3	2.0	3.0

Columns:

- **Node**: Node number
- **X**: Horizontal coordinate
- **Y**: Vertical coordinate

4.2 Link Data

The Link Data table defines the truss members.

Link	StartNode	EndNode	E	A
1	1	2	2.0×10^7	0.005
2	1	3	2.0×10^7	0.005
3	2	3	2.0×10^7	0.005

Columns:

- **Link**: Link or element number

- **StartNode**: Starting node of the link
- **EndNode**: Ending node of the link
- **E**: Young's modulus
- **A**: Cross-sectional area

4.3 Support Data

The Support Data table defines restrained and free degrees of freedom.

Node	FixX	FixY
1	Y	Y
2	N	Y

Use:

- `Y`: restrained degree of freedom
- `N`: free degree of freedom

4.4 Load Data

The Load Data table defines nodal loads.

Node	Fx	Fy
3	0.0	-10000.0

Sign convention:

- Positive `Fx`: load acts to the right
- Negative `Fx`: load acts to the left
- Positive `Fy`: load acts upward
- Negative `Fy`: load acts downward

5 Using the App

Step 1: Enter Input Data

Enter node data, link data, support data, and load data into the corresponding tables in the Input Panel. **Use consistent units throughout the model**. For example, if length is entered in metres and force in Newtons, then Young's modulus and area must also be entered in compatible units.

Step 2: Verify the Truss

Click the **Verify** button. The app checks the input data and plots the verification geometry.

The verification stage checks for:

- Empty node or link tables
- Duplicate node numbers
- Invalid start or end nodes
- Zero-length links
- Invalid support or load node numbers

If any error is detected or if the displayed verification geometry does not appear correct, return to the relevant input table and edit the data as required. Alternatively, the **Clear** button can be used to reset the input tables, allowing the truss data to be entered again from the beginning.

Step 3: Analyse the Truss

Click the **Analyse** button. The app solves the truss and displays the results.

The following outputs are generated:

- Nodal displacements
- Link axial forces
- Support reactions
- Deformed truss plot

Step 4: Review Results

Review the displacement, axial force, and reaction tables in the Results Panel. The final plot shows the deformed truss shape with an axial force contour. Positive axial force indicates tension, while negative axial force indicates compression.

Step 5: Export PDF Report

Click the **Export PDF Report** button to generate an A4 PDF report.

The report includes:

- Verification geometry

- Input tables
- Deformed truss plot with axial force contours
- Result tables

Wait for the pop-up message saying **“PDF report exported successfully.”** before opening the PDF file.

6 Output Results

6.1 Displacement

The nodal displacement table contains:

Node	Ux	Uy

where:

- **Ux**: horizontal nodal displacement
- **Uy**: vertical nodal displacement

6.2 Axial Force

The axial force table contains:

Link	Axial Force

Positive force indicates tension, and negative force indicates compression.

6.3 Reaction Force

The reaction table contains:

Node	Rx	Ry

where:

- **Rx**: horizontal reaction force
- **Ry**: vertical reaction force

7 Assumptions and Limitations

AnsyTruss-2D is developed for the static analysis of two-dimensional, pin-jointed truss structures. The following assumptions are adopted:

- The structure is a 2D pin-jointed truss and is in static equilibrium.
- Members carry axial force only; bending moments, shear forces, and torsional effects are not considered.
- Material behaviour is linear elastic.
- Deformations are assumed to be small.
- Loads are applied only at nodes.
- Connections are ideal frictionless pins.
- Member self-weight is not included unless converted into equivalent nodal loads.
- Geometric nonlinearity, material nonlinearity, buckling, and dynamic effects are not considered.
- Units must be consistent throughout the input data.

8 Troubleshooting

Problem	**Possible Solution**
Verification fails	Check that node numbers are unique and that all link start/end nodes exist.
Analysis does not run	Check supports, loads, and ensure the truss is stable.
No load arrow appears	Check that the load node exists and that Fx or Fy is not zero.
Unexpected displacement values	Check units for E, A, length, and force.
PDF does not open correctly	Wait for the export-success pop-up before opening the file.
Link table overflows in report	Use compact column values or reduce the number of visible rows.

9 Summary

AnsyTruss-2D provides a simple and structured workflow for defining, verifying, analysing, visualising, and reporting two-dimensional truss structures. As a MATLAB-based GUI tool, it supports static analysis of linear elastic pin-jointed trusses using the direct stiffness method. It is developed at Auckland University of Technology (AUT) as an educational and engineering calculation tool for truss analysis.